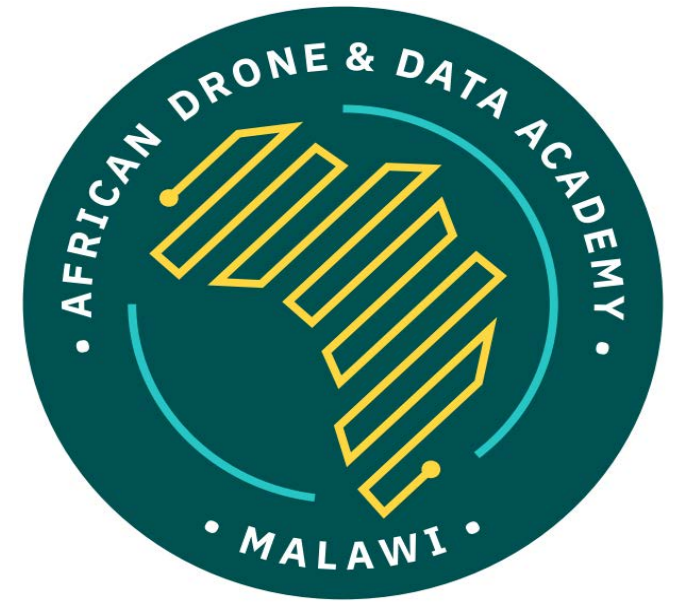




TOP Level 1

Day 6 Version A





TOP Level 1

Standard Operating Procedures Part 1



Standard operating procedures (SOPs) are used for assessments related to safety of flight and mission completion

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- **Project feasibility and safety assessments** are conducted using procedures and checklists that are established to ensure risk to the mission will be minimized
- **Pre-flight checklists** are used to force a critical look at several items in the flight operation with regard to function and safety
- **Non-normal scenarios and emergency procedures** are documented and classified to facilitate a rapid response if they occur
- **Post flight and pack-up** procedures allow the flight lessons to be recorded and support the next flight operation

Project feasibility and safety assessments

7. SOPs
8. SMS
9. Work safety
10. Crew roles

From the SMS:

- In all flight planning, multiple aspects of the flight operation must be considered to manage risk and maximize the success of the flight outcome
 - A flight planning process is used in *ADDAttracking* that guides a pilot through feasibility and safety assessment
 - A flight planning form (FP01) is used to enter initial data for the flight operation

Appendix 2: Initial Risk Assessment

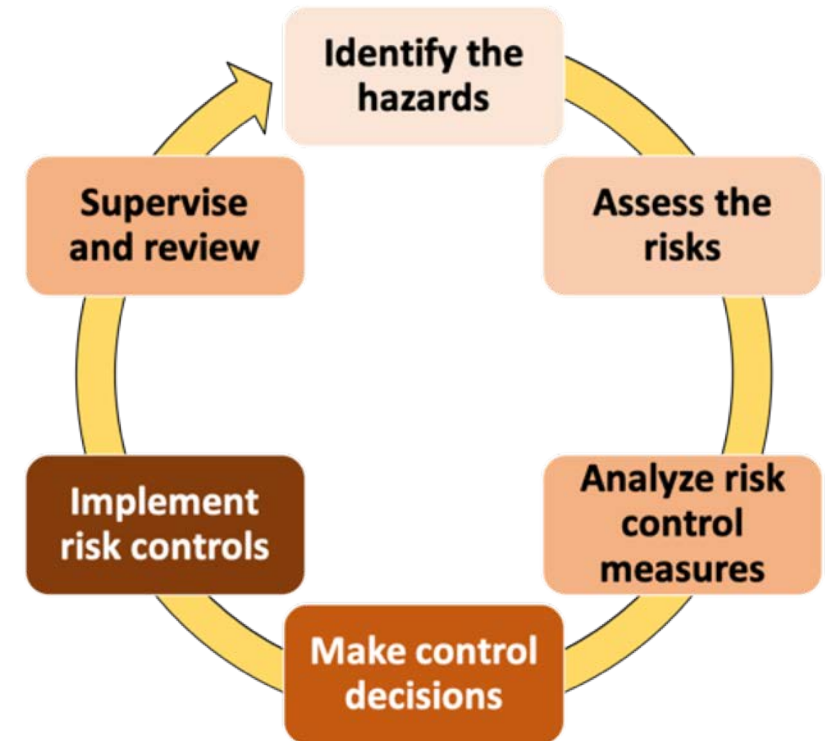
(Done for all new operations in Flight Request Form)

Initial Operations Risk Assessment Documentation		
Risk Assessment For: <i>Organization & Location</i>		Today's Date:
Reason for Risk Assessment (select all that apply):		
☐ New System Design	☐ Modification to an Existing Operation or Procedure	
☐ Change to Existing System Design	☐ Operational Environmental Change	
☐ New Operational Procedure	☐ Ineffective Risk Control	
☐ Other:		
Process/System Analysis		
Brief description of process of system to be assessed:		
Risk Analysis / Hazard Identification		
	Hazard	Potential Consequences
H1		
H2		
H3		

Risk management is conducted throughout a flight activity and it incorporates a **safety culture** starting with the preflight activity to post-flight debrief

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Using multiple checks throughout the flight operations process, we are able to manage risk effectively
- This means identifying risks and applying mitigations to reduce likelihood and severity



Project feasibility and safety assessments

7. SOPs
8. SMS
9. Work safety
10. Crew roles

From the SMS:

Appendix 5: Flight Risk Assessment Tool

(Implemented in Flight Request and Flight Planning Forms)

- A risk metric is computed in the flight planning process for all potential hazards
- An overall risk metric for the mission is determined (weighted scores on a 5-point scale)
- Mitigations are applied to control the risk to an acceptable level

		1	2	3	4	5	Rating
Operational Factors	Type of Operation	Normal	Demo	Training/ <u>Recurrency</u>	Initial Flight	Test Flight	
	Duration of Operation	< 1 hour	1-2 hours	2-4 hours	4-6 hours	>6 hours	
	Simultaneous Operations	1 UA		2 UAs		>2 UAs	
Crew Factors (any member)	Hours of Rest in Last 24 Hours	>8	7-8	5-6	3-5	<3 (No Fly)	
	# of Flights in Type	>20	10-20	5-10	1-5	0	
	# of Flights in Last 90 Days	>20	15-20	10-14	5-9	<5	
	Total UAS Hours	>20	10-20	5-10	1-5	0	
Environmental Factors	Current Wind or Max Gust	<8 <u>kts</u>	9-12 <u>kts</u>	13-15 <u>kts</u>	16-18 <u>kts</u>	>18 <u>kts</u> (No Fly)	

Project feasibility and safety assessments

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Although a 5-pt scale is the easiest to use when computing risk, a risk matrix provides a more accurate assessment of how risk can impact a mission
- The **risk matrix** considers two components: **likelihood** and **severity**

<u>Risk Likelihood</u>	<u>Risk Severity</u>				
	Catastrophic A	Critical B	Moderate C	Minor D	Negligible E
5 – Frequent	5A	5B	5C	5D	5E
4 – Likely	4A	4B	4C	4D	4E
3 – Occasional	3A	3B	3C	3D	3E
2 – Seldom	2A	2B	2C	2D	2E
1 – Improbable	1A	1B	1C	1D	1E

Project feasibility and safety assessments

- Severity is defined as the worst thing that can result due to an individual hazard (SMS Sec 3.2)
- Likelihood is defined as the degree to which the hazard is probable to occur (SMS Sec 3.2)

7. SOPs
8. SMS
9. Work safety
10. Crew roles

<u>Risk Likelihood</u>	<u>Risk Severity</u>		
	Catastrophic A	Critical B	Moderate C
5 – Frequent	5A	5B	5C
4 – Likely	4A	4B	4C
3 – Occasional	3A	3B	3C
2 – Seldom	2A	2B	2C
1 – Improbable	1A	1B	1C

Table 1. Safety risk severity table

Severity	Meaning	Value
Catastrophic	Equipment destroyed, death	A
Hazardous	Large reduction in safety margins, physical distress, serious injury, major equipment damage	B
Major	Significant reduction in safety margins, reduction in ability to cope with adverse conditions, serious incident, injury to persons	C
Minor	Nuisance, operating limitations, use of emergency procedures, minor incident	D
Negligible	Few consequences	E

Table 2. Safety risk probability table

Likelihood	Meaning	Value
Frequent	Likely to occur many times (has occurred frequently)	5
Occasional	Likely to occur sometimes (has occurred infrequently)	4
Remote	Unlikely to occur, but possible (has occurred rarely)	3
Improbable	Very unlikely to occur (not known to have occurred)	2
Extremely improbable	Almost inconceivable that the event will occur	1

Project feasibility and safety assessments

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- The outcome of the assessment of risk can be summarized below:

Assessment Risk Index	Criteria	Accountability
5A, 5B, 5C, 4A, 4B, 3A	Unacceptable under existing circumstances, requires immediate action.	Program Lead, Project Manager, Aviation Safety Officer, & Head of Training
5D, 5E, 4C, 3B, 3C, 2A, 2B	Manageable under risk control & mitigation. Requires authorized decision.	Project Manager, Aviation Safety Officer, & Head of Training
4D, 4E, 3D, 2C, 1A, 1B	Acceptable after review of the operation. Requires continued tracking and recorded action plans.	Project Manager or Aviation Safety Officer
3E, 2D, 2E, 1C, 1D, 1E	Acceptable with continued data collection and trending for continuous monitoring.	RPIC

An example of a risk analysis form is presented here

7. **SOPs**
8. SMS
9. Work safety
10. Crew roles

Risk matrix or “cube”

[illegible]

Example: \$50 bicycle

7. SOPs
8. SMS
9. Work safety
10. Crew roles

[illegible]

Example: \$50 bicycle

7. SOPs
8. SMS
9. Work safety
10. Crew roles

[illegible]

Example: \$50 bicycle

7. SOPs
8. SMS
9. Work safety
10. Crew roles

Risk Item No.:	3	Risk Identifier:	K. Kochersberger
Project Name:	\$50 Bicycle		
Project number:	10		
Risk Title:	Plastic wheel (2)		
Description:	The plastic wheel design cannot be adequately tested before injection molding tooling is made, resulting in a high technical risk if the system does not work. Other unknown problems include UV degradation, shatter failure, and lack of impact resistance. Assume that Risk Item 2 has already been mitigated.		
Basic Cause:	Injection molding bicycle wheels and wheel construction from plastics is a new and unproven technology. While analysis can prevent known problems from developing, there is the possibility of unknown problems causing the wheel to fail.		
Impact:	The wheel will not last long and will have to be replaced frequently, or the bicycle will not be used once the wheel becomes deformed beyond a usable tolerance		
Mitigation:	In addition to the injection molded plastic wheel design, an aluminum wheel concept will be pursued as a fallback design.		
Ratings:			
Likelihood	2		
Consequence	5		
Risk Level	Mod		
Risk Type:	T		
Potential Cost			
Potential \$ for Mitigation Items:			
Created:		date	
Last update:		date	
Last review:		date	
Status:		open	

All risks are then summarized in a risk table

7. SOPs
8. SMS
9. Work safety
10. Crew roles

Risk ID #	Risk Description	Likelihood	Consequence	Risk Level	Owner	Mitigation Plan	Mitigated Probability	Mitigated Consequence	Mitigated Risk Level	Status
1	The bamboo frame design may require specialized tooling to fabricate with adequate strength and straightness. If this is required, we will miss a target specification of fabrication without specialized tooling which compromises the ability to establish manufacturing in rural/undeveloped areas.	3	4	7	Stoepker	Plugs will be designed that fit into the bamboo and assist in aligning it to the mating components. A test will be conducted using the plug concept that validates the strength and straightness of the frame.	1	4	5	Pending
2	The plastic wheel design cannot be adequately tested before injection molding tooling is made, resulting in a high cost risk if the system does not work. Primary concerns are the possibility of plastic creep causing the wheel to deform over time.	4	5	9	??	A finite element analysis of the plastic wheel will be conducted to ensure the wheel stresses are below critical stresses for creep when subject to elevated temperatures.	2	5	7	Pending
3	The plastic wheel design cannot be adequately tested before injection molding tooling is made, resulting in a high technical risk if the system does not work. Other unknown problems include UV degradation, shatter failure, and lack of impact resistance. Assume that Risk Item 2 has already been mitigated.	2	5	7	??	In addition to the injection molded plastic wheel design, an aluminum wheel concept will be pursued as a fallback design.	1	3	4	Closed

Pre-flight checklists

- Checklists are used to make sure all critical elements of a system are inspected and verified for safe and reliable operation
 - *Items on a checklist are added when a critical component or system has been shown to contribute to failure if it is not configured properly*
 - In manned aircraft, many checklist items appear because an incorrect setting resulted in a *crash*
- There is a method for using a checklist:
 - **Read** the item on the list
 - **“Do”** the action, accompanied by pointing to or touching whatever is being checked
 - **Verify** that the action has been completed

7. SOPs
8. SMS
9. Work safety
10. Crew roles

<u>Checklist</u>	
☒	Item A
☒	Item B
☒	Item C
☐	Item D
☐	Item E

Pre-flight checklists

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- FOM 3.5.1: Aircraft-specific document review -
- For reference, all aircraft-specific documents are available on the ADDA Drive site. These documents are available for inspection
 - RPA operational certificate (ROC) or letter of approval (RLA), where applicable
 - Aircraft operations manual, where applicable, including operational limitations
 - Standard operating procedures manual (SOPM)
 - Maintenance manual, where applicable
 - Aircraft and battery logbooks
 - Certificate of registration, where applicable
 - Certificate of airworthiness, where applicable
 - Waivers of operation, if any

Pre-flight checklists: document review for aircraft

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- The aircraft logbook will contain a record of the history of maintenance and significant events regarding the operation of the aircraft
 - It is the best source of information to learn about the current operational state of the aircraft
- The **user manual** and **standard operating procedure manual (SOPM)** are **aircraft-specific documents**: they contain information about operations limitations and specific guidelines on operating the aircraft.



Pre-flight checklists: document review for aircraft

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Battery logs are important since batteries have a limited life and it is dependent both on their **age** and **number of charging cycles**.
- Check the battery logs frequently to make sure the batteries are capable of a successful flight before takeoff
 - Battery maintenance records should not show any irregularities that would indicate a failing battery
 - Signs of problems include a swelling of the battery pack and/or an inability to hold charge



Pre-flight checklists - document review for pilot

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- FOM 3.5.2: Document review - pilot
- The pilot needs to verify that their paperwork is up-to-date to be legal to operate:
 - Remotely piloted aircraft pilot's license (RPL) where applicable
 - Student pilot license, where applicable
 - Logbook review for flight currency, if applicable
 - Medical certificate, if applicable
 - Waivers to operate, if any



Pre-flight checklists - flight planning

7. SOPs
8. SMS
9. Work safety
10. Crew roles

The day-before checklist includes a site survey:

- **VTOL:**

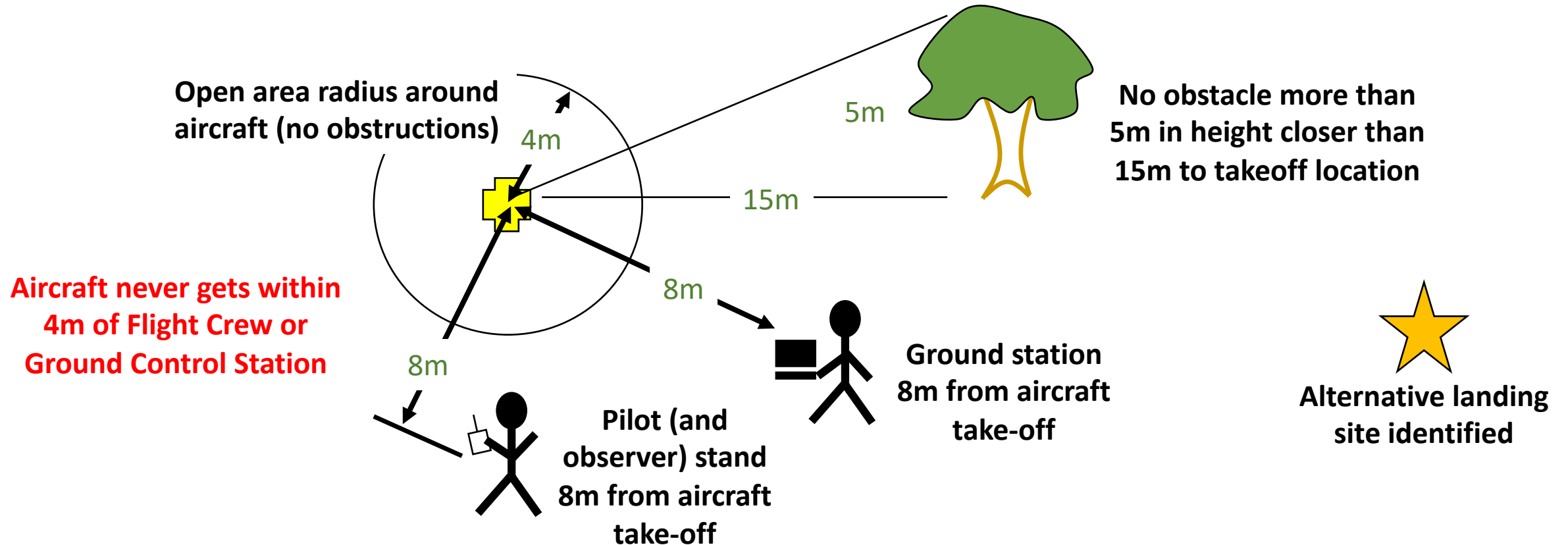
- Area is mostly level (low slope, minimal bumps)
- Open area with at least *4m* radius around the UAV
- Minimal loose debris to be kicked up during take-off or landing (place heavy blanket or plywood on ground if this is an issue)
- No vertical obstacles *above 5m* of flying altitude within *15m* radius of take-off location
- Pilot and observer location is at least *8m* feet from take-off point and have room to move from that location away from the UAV
- Pilot and observer have clear line of sight to UAV at take-off and can safely move to a position to maintain line of sight through entire flight. Beyond visual line of sight (BVLOS) flights are not allowed unless the operator has written permission from the agency controlling agency the airspace.
- Ground station is at least *8m* from take-off point

Pre-flight checklists - flight planning

The day-before checklist includes a site survey:

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- VTOL site dimensions:



Pre-flight checklists - flight planning

The day-before checklist includes a site survey:

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- **Fixed wing:**
 - Area is mostly level (low slope, minimal bumps)
 - Runway is long enough and wide enough for UAV takeoff and landing operations
 - No vertical obstacles greater than $5m$ high within a $35m$ perimeter of runway
 - Operator location is at least $8m$ from take-off point (start)
 - Operator has clear line-of-sight of UAV at take-off and can safely move to a position to maintain line of sight through entire flight. Beyond visual line of sight (BVLOS) flights are not allowed unless the operator has written permission from the agency controlling agency the airspace.
 - Ground station is at least $8m$ from take-off point

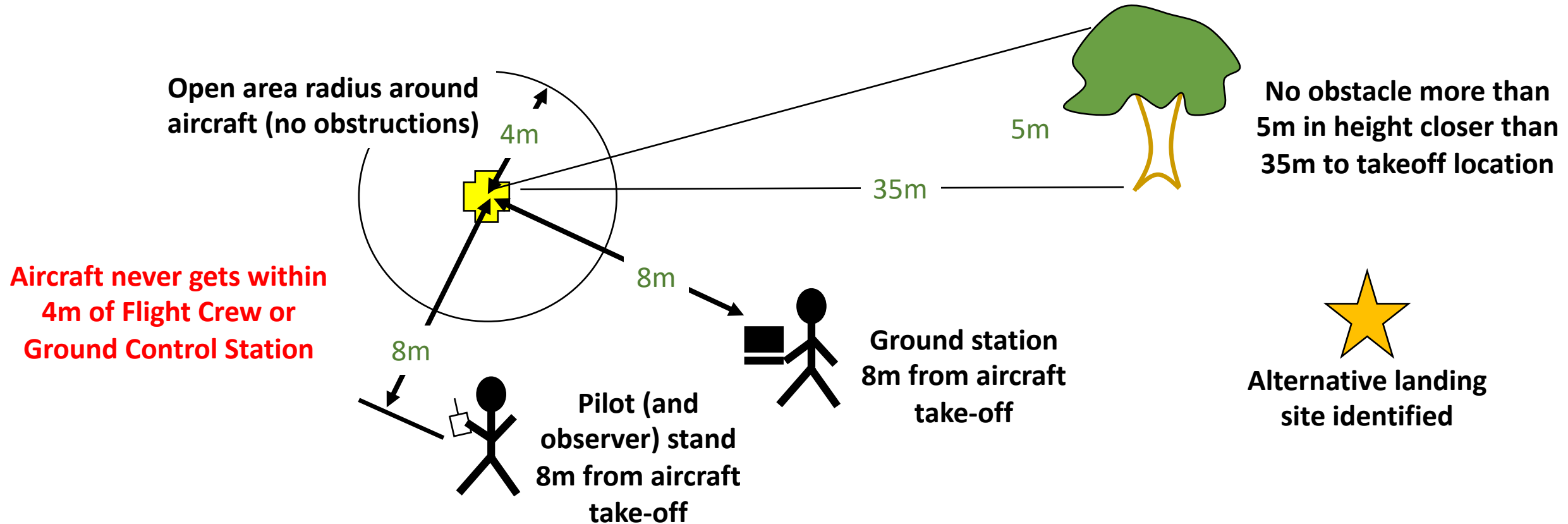


Pre-flight checklists - flight planning

The day-before checklist includes a site survey:

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Fixed wing site dimensions:





TOP Level 1

RPA Operations



Use and retention of the aircraft OEM handbook/manual

- The two aircraft-specific documents that are frequently referred to include the **user manual** and the **standard operating procedures manual (SOPM)**
 - The **user manual** contains details about aircraft function, maintenance, assembly and disassembly as well as specific information about the design and components used on the aircraft
 - The **SOPM** is focused on operation of the aircraft in the areas of flight activity that it was designed for

11. **RPA operations**
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Use and retention of the aircraft OEM handbook/manual

- Pilots are encouraged to become familiar with the user manual and the SOPM to operate the aircraft within its design parameters

S-500 Operating limitations:

- *Flight in any type of precipitation is prohibited.*
- *Surface wind limit – 15 knots.*
- *Winds aloft limit – 15 knots.*
- *The flight must be conducted in VFR conditions (surface visibility must be 1 SM and the cloud ceiling must be at least 1,000 ft.).*
- *The S500 must remain within VLOS of the PIC.*
- *Flights must occur within the altitude specifications set form in this document.*

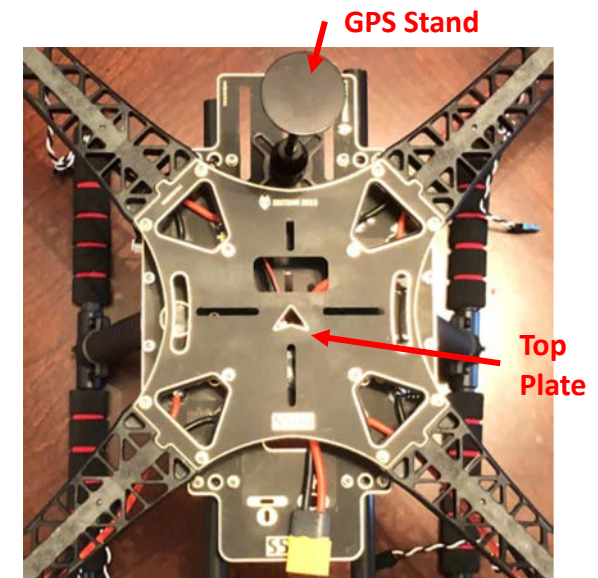
11. RPA operations
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Use and retention of the aircraft OEM handbook/manual

- Maintenance questions and procedures for aircraft assembly / conducting repairs are answered in the user manual
 - The specific instructions on how to assemble an aircraft and how to connect the wiring are provided in the user manual
 - Specific maintenance actions may not be called out, but the user manual will contain enough valuable information to proceed with operations required for aircraft servicing

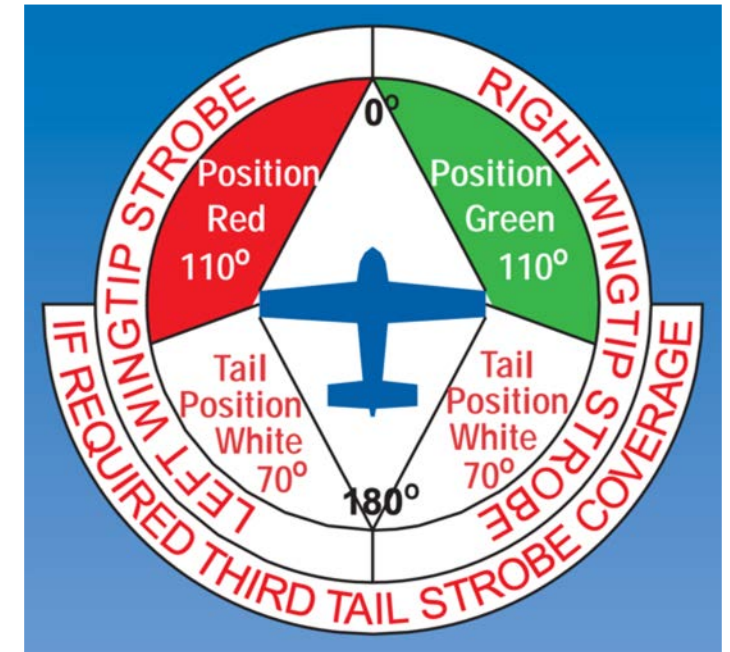
11. RPA operations
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Aircraft orientation and lighting

- Aircraft are marked using red, green and white lights that are installed around the extreme points of the aircraft
- These are used for identification during night flight
 - *Defined as the time between sunset and sunrise*
- The aircraft orientation can be interpreted by knowing the color of these lights in relation to their installation on the aircraft
- “Red Right Returning” means that if you see a red light on the right side, the aircraft is flying towards you

11. RPA operations
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Whelen strobe company

Aircraft orientation and lighting

- It may be to your advantage to mark the aircraft with asymmetric colors for orientation reference
 - This will help as the aircraft gets further away and orientation is difficult to observe
- If orientation is lost, the aircraft can be pitched forward and if it moves to the left, turn left to return
- If the aircraft moves right when pitched forward, turn right to return



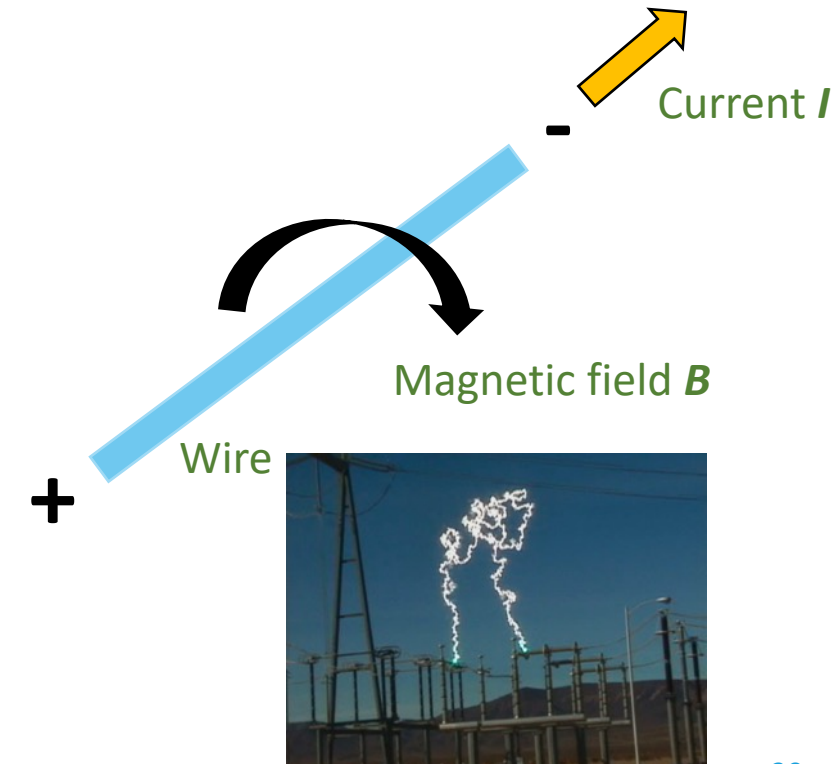
11. **RPA operations**
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Effects of EMI and RFI

- **EMI is electro-magnetic Interference, and RFI is radio-frequency interference**
- EMI and RFI may be used interchangeably
 - They are caused by electrical discharge, even from static electricity from clothing
 - They are also caused by short circuiting, or incomplete electrical connections that cause an arcing and sparking to occur
 - Another source of EMI/RFI is from high current in wiring that results in a magnetic field being generated in the vicinity of the wire
 - The magnetic field can cause other secondary electrical interference

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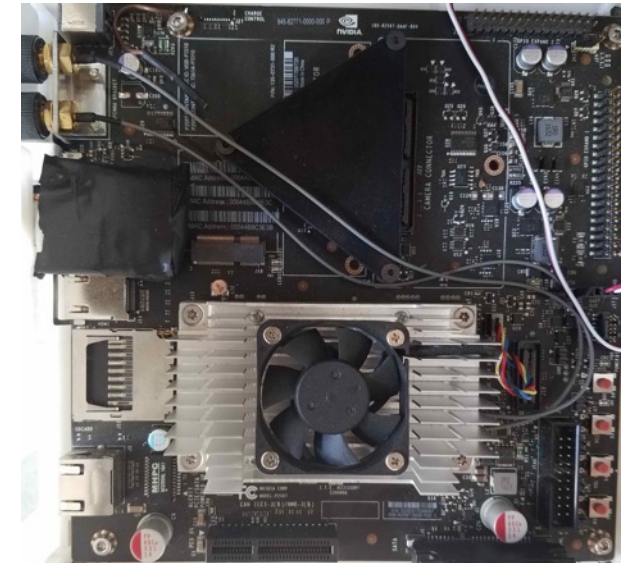


Effects of EMI and RFI

- **A few precautions are taken to reduce the likelihood of EMI / RFI - induced interference:**
 - USB-3 cameras are inherently very noisy and should be avoided
 - Single board computers are noisy and should be shielded
 - Maximize the distance between components if you cannot shield them independently
 - Twisted pair wires are used to reduce electromagnetic induction - a cause of magnetic interference which can induce electrical interference
 - Keep wires isolated as much as possible



USB-3 board camera



Single board computer

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19. Proficiency 2

Aircraft component care

- Aircraft are unique systems that must be optimized around payload capacity and endurance
- This requires that the aircraft structure be lightweight and strong
 - Light and strong aerostructures are **strong** in the load carrying direction and function, but they may be **weak** in other loaded directions
 - Put another way, the aircraft are only strong in the directions that they need to be strong while in flight; in other loaded directions, they may be fragile and will **require care in handling**

11. RPA operations
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Gossamer Condor in flight www.fai.org

Aircraft component care

- Propellers require careful inspection since a failure can be catastrophic for the aircraft
- They are strong to carry distributed loads, but not strong with concentrated loading
- In the event of a prop strike, propellers should be thoroughly inspected for delamination
- A nick on the leading edge can create a stress concentration that propagates to a much larger failure condition
- A “tap” test can be used to check the propeller for delamination damage
 - Tap with a coin, listen for a change in pitch along the length of the blade
 - **Note that delamination can be difficult to find - inspect carefully!**

11. RPA operations
12. Flight performance
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19. Proficiency 2



T-motor propellers - carbon epoxy with a foam core. They are precisely finished and have high efficiency, but can delaminate in a prop strike causing failure

Aircraft component care

- The landing gear on fixed wing and VTOL aircraft is usually the first part of the aircraft to contact the ground in a hard landing
- As a result, always inspect the landing gear for damage before and after flying
 - Cracks in plastic and carbon epoxy components can be difficult to see, requiring a close-up inspection
 - Note that landing gear components on aircraft are frequently under-designed - they always deserve a good look before flight!

11. RPA operations
12. Flight performance
13. Meteorology
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Aircraft component care

- Payloads also represent critical components of the aircraft system
 - In some cases, if they are not protected from damage during transport they should be removed from the aircraft
- Always consider ground handling damage to be a primary cause of component failure
 - Loading is frequently applied that the aircraft was not designed to withstand
 - When carrying VTOL aircraft, carry them by the landing gear or rotor arms - these are strong parts of the aircraft and are safe for handling

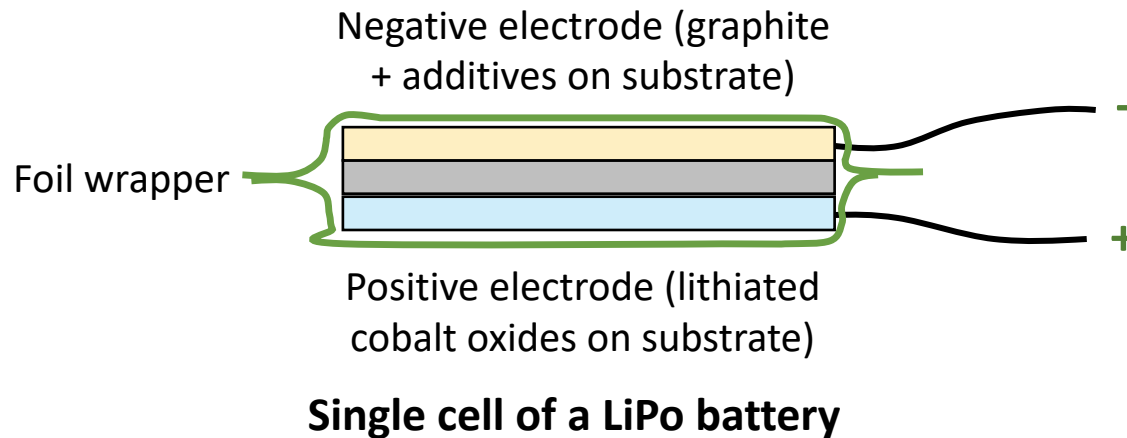
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Battery care: management, storage and transportation

- Battery technology today is significantly better than just a few years ago, however with the advances in performance comes some risks
- The common battery used in drones is a Lithium Polymer (LiPo) battery design

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19. Proficiency 2



Each cell of a LiPo battery is contained in a soft aluminum pouch, allowing for battery expansion during charge and discharge cycles (up to 10% dimensional change)



A single LiPo battery cell has a nominal voltage of 3.7V

Battery care: management, storage and transportation

- Battery voltages are important to regulate as they can impact the life of the battery
 - A single LiPo cell will have a nominal (average voltage) of **3.7V (50% discharge)**
 - The lowest voltage that battery should experience is **3.0V**, and the highest voltage from charging should be **4.2V**
 - A battery rated at **7.4V** is comprised of **two cells (2) wired in series (S)**
 - A **2S** battery has two cells in series (**7.4V**), a **3S** battery has 3 cells in series (**11.1V**)

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Protection electronics are required with these batteries!

- *To avoid charging above the max value*
- *To avoid discharge below min value*
- *To avoid excessive discharge*

Battery care: management, storage and transportation

- Charging systems typically have built-in safeguards to protect the battery:
 - Overload, overvoltage and temperature protections are common
 - Cell balancing avoids an unstable charging process which could lead to overheating - this is accomplished through monitoring by connecting to the balancing leads on the battery
- The charger output needs to be matched to the battery type (2S, etc)
- All battery charging stations should be operated in fire-safe environments because the risk of battery fire is elevated during charging

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19. Proficiency 2



Battery care: management, storage and transportation

- Fire safety is critically important when LiPo batteries are being used
 - Charging should not occur unattended - ***no overnight charging!***
 - Batteries that become “puffy” (greater than 10% dimensional change) are probably damaged due to improper charging/discharging. ***Take them out of service***
 - Batteries should be stored in fireproof containers or at a minimum, in non-combustible areas
 - A container of sand should be near the battery charging station to put fires out
 - **Water is not used on a battery (chemical) fire**

11. RPA operations
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19. Proficiency 2



<http://instructables.com>

Battery care: management, storage and transportation

- Similar to storage, the LiPo battery should be transported in a fireproof container, or at the least, kept away from flammable articles
- The leads on a LiPo battery should be protected when not in use to minimize the likelihood of a short circuit
- Batteries should not be transported in close proximity to each other - if one overheats it may cause the others to spontaneously overheat as well
- Always keep a multimeter or cell checker nearby to check the state of charge of the battery

11. RPA operations
12. Flight performance
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19. Proficiency 2

